INCH-POUND

A-A-20196
August 2, 1993

COMMERCIAL ITEM DESCRIPTION

TOPPING, DESSERT AND BAKERY PRODUCTS, FROZEN OR DEHYDRATED

The U.S. Department of Agriculture has authorized the use of this Commercial Item Description in preference to Military Specification MIL-T-43856A.

This Commercial Item Description (CID) covers frozen topping and dehydrated topping (requiring reconstitution) for dessert and bakery products, packed in commercially acceptable containers, suitable for use by the Federal Government.

Salient characteristics.

The product shall conform to the following types and styles, as specified in the solicitation, contract, or purchase order.

Types and styles.

Type I - Frozen

Style A - With nonfat milk solids
Style B - Without nonfat milk solids

Type II - Dehydrated

Type I - Frozen topping shall be prepared from refined deodorized vegetable fat or oil which may contain antioxidants (of the kinds and levels permitted in accordance with the Food and Drug Administration (FDA) Food Additive Amendment concerning rendered animal or processed vegetable fats) and emulsifying agents, skim milk or U.S. Extra Grade (7 CFR Part 58) spray process low-heat nonfat dry milk (for use in Style A), or food grade protein (for use in Style B), nutritive carbohydrate sweeteners, vanilla flavoring (natural and/or artificial), artificial color, and may contain salt. The

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FSC 8940

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ingredients (except for flavoring) shall be blended, pasteurized and homogenized (or vice versa), cooled, and packaged. The vanilla flavoring shall be added to the cooled product prior to packaging. The product shall be frozen immediately after packaging. The thawed product shall have a uniform white to light cream color, a uniform creamy consistency, and shall be characteristically sweet, with a clean sweet and pleasing vanilla flavor, free from stale, rancid, or oxidized flavors.

Type II - Dehydrated topping shall be prepared from refined deodorized vegetable fat or oil which may contain antioxidants (of the kinds and levels permitted in accordance with the FDA Food Additive Amendment concerning rendered animal or processed vegetable fats) and emulsifying agents, skim milk or U.S. Extra Grade (7 CFR Part 58) spray process low-heat nonfat dry milk, food grade protein, nutritive carbohydrate sweeteners, vanilla flavoring (natural and/or artificial), artificial color, and may contain salt and anticaking agents. The ingredients (except for the flavoring) shall be blended, pasteurized and homogenized (or vice versa), cooled, and spray dried. The vanilla flavoring may be added to the cooled product prior to spray drying or a powdered vanilla flavoring may be dry blended with the spray dried product. The dehydrated topping shall be light cream in color and free of lumps which do not fall apart under light finger pressure.

Analytical requirements.	Type I	Type II
Total fat, not less than (percent)	20	34
Total solids, not less than (percent)	50	--
Moisture, not more than (percent)	--	1.50
Scorched particle content per 32.5 grams of product, not more than (mg)	--	15.0
Phosphatase activity (shall be tested when skim milk is used in the formulation)	Less than 1 ug of phenol per mL of product	

Microbiological requirements.

The topping shall have a standard plate count of not more than 10,000 per gram and a coliform count of not more than 10 per gram. The salmonella test shall be negative.

Physical requirements.

The whipped product shall be white to light cream in color, have a stiff body, dry appearance, and a smooth creamy texture. The whipped product shall not be frothy or fluffy, nor break down in the emulsion when whipped (within a 10 minute limitation), shall be uniform in composition, and shall be characteristically sweet with a vanilla flavor background and free from objectionable flavors and odors. The Type I (after defrosting) and Type II

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(after reconstitution) toppings shall whip to a minimum 160 percent overrun after no more than 10 minutes of high speed whipping and not show serum separation nor weeping when held 2 hours at ambient temperatures of 70 to 80°F (21.1 to 26.7°C).

Test methods.

The following chemical, microbiological, and physical analyses shall be made in accordance with the following referenced and specified methods. The Type I samples shall be thoroughly defrosted and tempered to 40°F (4.4°C) prior to all sampling and testing. The Type II product shall be thoroughly blended before testing, in the event stratification has occurred during shipment.

Tests.

Official Methods of Analysis of the AOAC

Test	Method number Type I	Method number Type II
Fat	952.06	932.06
Total solids	941.08	--
Moisture	--	927.05
Salmonella 1/	967.26	967.26
Standard plate count	966.23	966.23
Coliform count	966.24	966.24

1/ For Salmonella testing, proceed as for dried egg, except using nutrient broth instead of lactose broth in initial dilution.

Standard Methods for the Examination of Dairy Products

Type I

Phosphatase (when skim milk is used in the formulation)	Chapter: Phosphatase Methods Section: Scharer Rapid Phosphatase Test (follow method for cream)

DA Instruction No. 918-RL, Instructions for Resident Grading Quality Control Service Programs and Laboratory Analysis

Type II

Scorched particle content	32.5 grams of dry product using the procedure for whole milk

Whipping test. The whipping test is conducted to determine the overrun potential and serum separation of the product. The whipping test should be

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run using a Hobart model N-50 or equivalent, having speeds ranging from about 60-260 RPM of planetary rotation and 135-600 RPM of agitator rotation, and with an approximately 3-quart (2.84 L) capacity bowl. For the Type II product, reconstitute in accordance with the manufacturer's instructions except that the product shall be gently mixed with a spoon or spatula to completely incorporate the dry powder into the liquid, being careful not to whip air into the product. The product shall be tempered to 40 to 45°F (4.4 to 7.2°C) before conducting the whipping test.

1. Flush fill a tared open-top 1-pint (0.47 L) container with the liquid (thawed Type I or reconstituted Type II) topping base. Record the weight. Pour product into a chilled [40 to 45°F (4.4 to 7.2°C)] whipping bowl with the balance of the product.

2. Start whipping at a low speed (No. 1) to prevent splashing. When thickening is observed, increase to a medium whipping speed (No. 2). When a noticeable volume increase is obtained, finish whipping at high speed (No. 3) until standing peaks are formed. High speed whipping shall not exceed 10 minutes. Record the whipping time.

3. Carefully transfer the whipped topping into a tared open-top 1-pint (0.47 L) container. Special care must be taken to eliminate air voids, (avoid expelling air by pressing down with a spoon or spatula). Strike off excess and wipe off product spilled on the outside of the container, weigh, and record net weight of topping.

4. Calculate the overrun (OR) by the formula below. If the minimum required 160 percent OR has not been obtained and the high speed whipping time is less than 10 minutes, then return the product to the bowl and whip at high speed for the balance of the 10 minutes, then recalculate the OR.

$$\text{percent OR} = \frac{\text{weight of 1 pint (0.47 L) liquid mix - weight of 1 pint (0.47 L) whipped topping}}{\text{weight of 1 pint (0.47 L) whipped topping}} \times 100$$

5. The whipping test shall be run on one sample in duplicate, and the results of the two tests averaged. If the individual results differ by more than 25 percent, the whipping test shall be repeated once more, and all three results averaged and reported.

6. For serum separation testing, flush fill an 8-ounce (236.59 mL) container with the whipped topping, avoiding air pockets. Transfer the whipped topping onto a 6-inch square (38.71 cm^2) of (No. 26) B&S 16 gauge woven wire mesh which is resting on top of an appropriate sized support funnel and suspended over a collecting vessel. The product shall be held 2 hours at 70 to 80°F (21.1 to 26.7°C), showing no evidence of serum separation.

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Contractor's certification.

By submitting an offer, the contractor certifies that the product offered meets the specified salient characteristics and requirements of this CID; conforms to the producer's own specifications and standards, including product characteristics, manufacturing procedures, quality control procedures, and storage and handling practices; has a national or regional distribution from storage facilities located within the United States, its territories, or possessions; and is sold on the commercial market. The Government reserves the right to determine proof of such conformance prior to the first delivery from point of origin and any time thereafter, up to and including delivery at final destination, as may be necessary to determine conformance with the provisions of the contract.

Regulatory requirements.

The delivered product shall comply with all applicable Federal and State mandatory requirements and regulations relating to the preparation, packaging, labeling, storage, distribution, and sales of the product within the commercial marketplace. Delivered product shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

Quality assurance.

Prior to award of contract, the plant which manufactures the topping and the plant which packages the topping shall be approved by the Dairy Grading Branch, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture. Approved plants must satisfactorily meet the requirements contained in the "General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service" (7 CFR Part 58) and shall be eligible for listing in the publication, "Dairy Plants Surveyed and Approved for USDA Grading Service". When required in the solicitation, contract, or purchase order, the Dairy Grading Branch will determine the quality of the topping, according to applicable standards and Dairy Grading Branch procedures. The product shall be examined or analyzed in accordance with applicable provisions in the CID, and when applicable, the United States Standards for Condition of Food Containers currently in effect on the date of solicitation.

Preservation, packaging, packing, labeling, and marking.

The topping shall be preserved, packaged, packed, labeled, and cases marked in accordance with good commercial practice. Commercial labeling and packaging, as may be augmented by the solicitation, contract, or purchase order, shall be acceptable. Shipping containers shall comply with the National Motor Freight Classification or Uniform Freight Classification, as applicable.

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For Department of Defense procurements (Only).

The following requirements are applicable when specified by the contracting officer.

A. Commercial packaging. Product shall be packaged in accordance with good commercial practice in the following weights: 6-1/2 or 7 pounds (2.948 kg or 3.175 kg) of Type I product shall be filled into a size 603 x 700 open-top style, round, metal can. One pound (0.454 kg) of Type II product shall be packaged in a size 401 x 411, 401 x 509, or 401 x 510 open-top style, round, metal can.

B. Commercial packing. Six cans of Type I product, or twenty-four cans of Type II product shall be packed in containers complying with the National Motor Freight Classification or Uniform Freight Classification, as applicable.

C. Export packaging. Six and one-half or seven pounds (2.948 kg or 3.175 kg) of Type I product or one pound (0.454 kg) of Type II product shall be unit packaged as specified above. Canned product for export shall be in open-top style, round, metal cans with welded side seams and compound-lined, double-seamed ends. Cans shall be made throughout from not less than 0.25 pound (113.398 g) electrolytic tinplate per base box. The cans shall be coated overall outside with a coating conforming to Type I or III of TT-C-495. The cans shall be hermetically sealed and leak free.

D. Export packing. Six 603 x 700 cans or twenty-four 401 x 411, 401 x 509, or 401 x 510 cans shall be packed in fiberboard boxes, constructed, closed, and reinforced in accordance with Style RSC, Grade V3c, V3s, or V4s of PPP-B-636. Reinforcement shall be restricted to nonmetallic strapping or pressure-sensitive adhesive, filament-reinforced tape in accordance with the appendix of PPP-B-636.

E. Unit loads (commercial and export). Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for Type II, Class F loads.

F. Labeling (commercial and export). Commercial labeling shall be acceptable.

G. Marking (commercial and export). Marking of shipping containers and unit loads shall be in accordance with MIL-STD-129. In addition, the following precautionary marking shall appear on each shipping container and each unit load of Type I product in letters not less than 1 inch high:

PERISHABLE - KEEP FROZEN
0°F OR BELOW

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Notes.

Purchasers shall specify product type(s) and style(s).

Sources of documents.

Sources of information for nongovernmental documents are as follows:

Copies of the National Motor Freight Classification may be obtained from:

National Motor Freight Traffic Association, Inc., Agent
National Motor Freight Classification
American Trucking Associations, Inc., Traffic Department
2200 Mill Road
Alexandria, VA 22314

Copies of the Uniform Freight Classification may be obtained from:

Uniform Classification Committee, Agent
Uniform Freight Classification
Uniform Classification Committee, Suite 1120
222 South Riverside Plaza
Chicago, IL 60606

Copies of the Official Methods of Analysis of the AOAC may be obtained from:

AOAC International
2200 Wilson Boulevard
Suite 400
Arlington, VA 22201-3301

Copies of the Standard Methods for the Examination of Dairy Products of the American Public Health Association may be obtained from:

American Public Health Association
1015 18th Street, NW
Washington, DC 20036

Sources of information for governmental documents are as follows:

Applicable provisions of the Federal Food, Drug, and Cosmetic Act are contained in 21 CFR Parts 1 to 199. This three-volume set may be purchased from:

Superintendent of Documents
U.S. Government Printing Office
Washington, DC 20402-0001

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Credit card (MasterCard or Visa) purchases may be made by calling the Superintendent of Documents on (202) 783-3238.

Copies of "General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service," "Dairy Plants Surveyed and Approved for USDA Grading Service," "United States Standards for Grades of Nonfat Dry Milk (Spray Process)," and "DA Instruction No. 918-RL, Instructions for Resident Grading Quality Control Service Programs and Laboratory Analysis" are available from:

Dairy Standardization Branch
Dairy Division
Agricultural Marketing Service
U.S. Department of Agriculture
Room 2750, South Building
P.O. Box 96456
Washington, DC 20090-6456

Copies of the United States Standards for Condition of Food Containers are available from:

Chairperson
Condition of Container Committee
Agricultural Marketing Service
U.S. Department of Agriculture
Room 2506, South Building
P.O. Box 96456
Washington, DC 20090-6456

Civil agencies and other interested parties may obtain copies of this CID from:

General Services Administration
Specifications Unit (3FB-WS)
Room 6654
7th and D Streets, SW
Washington, DC 20407

Military activities should submit requests for copies of this CID to:

Standardization Documents Order Desk
Building 4, Section D
700 Robbins Avenue
Philadelphia, PA 19111-5094

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MILITARY INTERESTS:

Custodians

Army - GL
Navy - SA
Air Force - 50

Review Activities

Army - MD, QM
Navy - MC
DLA - SS

CIVIL AGENCY COORDINATING ACTIVITIES:

DOJ - BOP
HHS - FDA, NIH
USDA - FV
VA - OSS

PREPARING ACTIVITY:

Army - GL

Project 8940-0734

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